Division of Engineering Science
University of Toronto
MIE286 H1 S: Probability & Statistics

Course Instructor: Birsen Donmez
LEC01: Mon 11-12pm SF3202; Wed 1-2pm MS2172; Fri 11-12pm MC252
LEC02: Tues 3-4 pm BA1130; Wed 2-3 pm MC252; Fri 3-4pm MC252
E-mail: donmez@mie.utoronto.ca
Office hours (starts the week of Jan 12th): Tues 2-3pm & Fri 1-2pm; Online on Zoom, or in person at RS207

Teaching Assistants: ***Head TA:*** Koorosh Moslemi (koorosh.moslemi@mail.utoronto.ca)

Tutorial TAs:
Fan Wang (fanw.wang@mail.utoronto.ca) – ***Project***
Tianyi (Sherry) Mao (sherry.mao@mail.utoronto.ca) – ***Project***
Linzhuo (Mark) Wei (mwei@mie.utoronto.ca)
Hshmat Sahak (hshmat.sahak@mail.utoronto.ca)

Course Website: https://q.utoronto.ca/

Required Textbook: R.E. Walpole, R.H. Myers, S.L. Myers, and K. Ye. *Probability & Statistics For Engineers & Scientists*, Ninth Edition, Prentice Hall, 2011

Required Software: R [can be downloaded for free at http://www.r-project.org/]; RStudio [can be downloaded for free at https://posit.co/downloads/]

Lectures: 3 hours per week
Reading Week and Family Day: Feb 16th – Feb 20th
No class or office hours on Jan 23rd (Annual Engineering Science Education Conference) and Apr 3rd (Good Friday)

Tutorials: 1 hour per week

Course Overview and Learning Objectives: A course in probability and statistics for Engineering Science students focusing on building solid probabilistic and statistical foundations both mathematically and in terms of engineering application. Topics include sample space, events, definitions of probability, conditional probability, Bayes' theorem, important classes of discrete and continuous random variables and their distributions, joint, conditional, and marginal distributions, expectation, moment generating and characteristic functions, transformations of random variables, central limit theorem and approximations. Graphical methods, quantile plots, point and interval estimation of population parameters, method of maximum likelihood. Hypothesis testing, simple and multiple regression, correlation analysis, and introduction to Bayesian statistics.

Evaluation:

Quizzes during Tutorials	6%
Term Test to be held on Thursday, Mar 5, 9-11 am, EX320, MS2173, MS3154	30%
Project (Due TBD)	19%
Final Exam (D type)	45%

Tutorials and Quizzes

Tutorials start on the week of **January 12** and end the week of **March 30**. When there is a quiz, you should attend the tutorial section you registered for on ROSI.

Tutorial	Time	Location	TA
TUT01	Tues 12-1 pm	BA2139	Sherry
TUT02	Tues 12-1 pm	BAB025	Hshmat
TUT03	Th 1-2 pm	HA410	Mark
TUT04	Th 1-2 pm	BA1240	Fan
TUT05	Mon 2-3 pm	BA2179	Mark
TUT06	Mon 2-3 pm	BAB026	Hshmat
TUT07	Wed 3-4 pm	BA2179	Fan
TUT08	Wed 3-4 pm	BAB025	Sherry
TUT09	Mon 1-2 pm	BA2179	Mark
TUT10	Mon 1-2 pm	BA2165	Hshmat
TUT11	Wed 4-5 pm	BAB025	Sherry
TUT12	Wed 4-5 pm	BA2179	Fan

Practice Problems

Practice problems will be assigned weekly and will be posted on the website. They are **not** to be handed in. They will be appropriate preparation for the tutorial discussion and the quizzes.

Important Notes

- An 8 ½" x 11" **aid sheet** will be permitted for the term test (one sided). For the final exam, you can write on both sides. Write your name on your aid sheet. Only non-programmable calculators (Type 2) will be permitted for the quizzes, term test, and the final exam.
- All **term-work petitions must be submitted through the online petition system**, which is accessible through the Engineering Portal.
- Important announcements, additional examples, past tests/exams and other course info will be posted on the course webpage. **Check it regularly**!
- The instructor and TAs are here to help. Ask questions and alert us to any problems right away!

Academic Integrity

All students, faculty and staff are expected to follow the University's guidelines and policies on academic integrity. For students, this means following the standards of academic honesty when writing assignments, collaborating with fellow students, and writing tests and exams. Ensure that the work you submit for grading represents your own honest efforts. Plagiarism, representing someone else's work as your own or submitting work that you have previously submitted for marks in another class or program, is a serious offence that can result in sanctions. Speak to me or your TA for advice on anything that you find unclear. To learn more about how to cite and use source material appropriately and for other writing support, see the U of T writing support website (https://writing.utoronto.ca/). For more information, please see the U of T Academic Integrity website (https://www.academicintegrity.utoronto.ca/).

Normally, students will be required to submit their course essays to the University's **plagiarism detection tool** for a review of textual similarity and detection of possible plagiarism. In doing so,

students will allow their essays to be included as source documents in the tool's reference database, where they will be used solely for the purpose of detecting plagiarism. The terms that apply to the University's use of this tool are described on the Centre for Teaching Support & Innovation web site (https://uoft.me/pdt-faq).

Use of Generative AI Tools

The ability to communicate engineering knowledge effectively across a range of contexts and audiences is a key learning objective in this course. Your submitted assignments must demonstrate this learning objective, and as such, should be your own original and independent work. Any work submitted with the direct or indirect use of AI tools must be acknowledged in a form appropriate to the use, and failing to do so may result in academic offence.

Because of this, generative AI tools (such as, but not limited to ChatGPT) may not be used to create text for use in your reports. However, generative AI tools may serve as a tool for parts of the writing process, such as brainstorming, outlining, and research, but must be acknowledged appropriately. When acknowledging your use of Generative AI tools, please do the following: provide a citation in your works cited using the following format: “Tool Name, URL, Date used, Description how it was used, Prompt(s) used."

Since similar prompts will generate different responses at different times, if you use generative AI as a source, you must include a screenshot of the output in an appropriately titled appendix.

Use of direct quotes from Generative AI is not allowed, as it is not considered your independent work and Generative AI tools do not carry any inherent authority. If you have any questions about the use of generative AI in your assignments, please contact your instructor.

Discussion on Piazza

In addition to the tutorials and instructor office hours, we will use Piazza (www.piazza.com) for class-related discussions. We encourage you to ask questions on Piazza when you’re struggling to understand a concept—you can even do so anonymously (not to the instructors but to your peers). Questions of general interest to the class can be posted on Piazza and will be monitored and answered by the teaching staff. Further, students can also answer each other’s questions.

In general, your individual questions or issues should be addressed through your TA. If your TAs are unable to answer a question or resolve a problem, they will refer it to the instructor. If you wish to not use Piazza for class-related questions, you can ask your questions to your TAs during the tutorials or come to instructor office hours.

Piazza class site: https://piazza.com/utoronto.ca/winter2026/mie286h1slec0101.

Tentative Work Schedule

***Quiz weeks:** Jan 19, Feb 9, Mar 23

Week	Topic	Reading
1 Jan 5	Introduction to statistics and data analysis, the role of probability. Random sample, measures of location and variability. Graphical methods and data description. Histogram and dot-plot. Sample space and events. Definitions of probability. Conditional probability and the Bayes' rule.	Chapter 1, Chapter 2, sections 2.1, 2.2, 2.4 - 2.8.
2-3 Jan 12 Jan 19*	Concept of a random variable. Discrete and continuous probability distributions. Joint, marginal, and conditional distributions. Statistical independence. Mathematical expectation, variance, and covariance of random variables. Correlation coefficient.	Chapter 3, Chapter 4, sections 4.1, 4.2, 4.3
4-5 Jan 26 Feb 2	Discrete probability distributions: discrete uniform, binomial and multinomial distributions, hypergeometric, negative binomial, and geometric distributions, Poisson distribution. Poisson approximation to binomial. Homogeneous Poisson process. Continuous probability distributions: continuous uniform, normal distribution and its applications, exponential and gamma distributions, and Weibull distributions, the failure rate. Applications in reliability.	Chapter 5, Chapter 6, sections 6.1 – 6.6, 6.10, 6.11.
6 Feb 9*	Transformations of discrete and continuous random variables. Moment generating function. Chi-squared distribution.	Chapter 6, section 6.7. Chapter 7.
7-8 Feb 23 Mar 2	Data display and graphical methods. Quantiles, quartiles and percentiles. Sampling distributions, Central limit theorem and approximations with continuity correction for discrete distributions. Applications. Chebyshev's and Markov's theorems. The law of large numbers. Sampling distributions of $\bar{X}$ and S^2. Distributions arising from the normal: chi-squared, t, and F distributions.	Chapter 8. Chapter 4, section 4.4
9 Mar 9	Estimation of population parameters. Point and interval estimators. Unbiased and most efficient estimator. Standard error of a point estimate. Confidence and prediction interval. Absolute and relative error, sample size calculation. Single and two sample estimators. Method of maximum likelihood.	Chapter 9.
10 Mar 16	Hypothesis testing. Null and alternative hypotheses, type I and type II errors. Significance level and P-value, power of the test. Tests on means and variances for a single and two samples. Choice of sample size. Fitting a distribution to the data, Goodness of fit tests. Applications.	Chapter 10, sections 10.1-10.14
11 Mar 23*	Simple linear regression and correlation. Least squares estimation and model fitting. Inferences for regression coefficients. Data with repeated observations. Residual analysis and model checking. Prediction. Correlation analysis.	Chapter 11.
12-13 Mar 30 Apr 6	Multiple linear regression. Model fitting. Inferences in multiple regression. Model checking, criteria for model selection. Sequential methods for model selection. Residual analysis. Applications.	Chapter 12 (selected sections TBD)